

Structured Chunking for Enterprise Search

Abstract. We study how the layout of business documents affects retrieval. Our pipeline keeps every character of a document addressable, so answers cite the exact page and box.

1 Introduction

Retrieval quality depends on how documents are split. Chunks that respect the structure of the source keep headings, lists and tables together, so a passage that answers a question is found as one unit instead of three fragments. Retrieval quality depends on how documents are split. Chunks that respect the structure of the source keep headings, lists and tables together, so a passage that answers a question is found as one unit instead of three fragments.

Three properties matter for a knowledge base:

- every chunk maps back to a page region;
- unchanged paragraphs are never embedded twice;
- deletes leave no trace in any index.

2 Method

2.1 Layout analysis

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Pages are read column by column. A paragraph that reaches the bottom of a column and continues at the top of the next one is joined again, so its sentence is not cut in half by the layout of the page and the reader of the chunk sees the whole statement that the author wrote, including the clause that ends after the column break and the final words that close the sentence on the following column

2.2 Chunking

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The final paragraph of this section is long enough to run from the bottom of the second column of the first page onto the top of the first column of the second page, which tests the merge of a sentence across a page break as well as across columns, because readers expect the text to flow without interruption from one page to the next one

3 Results

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